COURSE GUIDE & SYLLABUS · GRADES 6–8 · AGES 11–14

Scratch (Gr 6–8) Beginner: Code Foundations

Algorithms, loops, logic and game loops — the big ideas of programming, built in Scratch.

A fast-paced first course for middle-school learners. Each module teaches a core computer-science idea — algorithms and events, iteration, variables and Boolean logic, and the game loop — through Code Lab challenges and a real project (Interactive Poster, Generative Art, Smart Assistant App, and an original Arcade Game capstone). Every lesson links the Scratch blocks to the equivalent Python, so learners are ready for text-based code. Each lesson has a video, a build-along, auto-checked Code Lab challenges, a worksheet and a teacher lesson plan.

What learners will be able to do

- Write, trace and debug precise algorithms
- Build event-driven programs with parallel scripts and broadcasts
- Use counted and forever loops, including nested loops
- Generalise solutions with variables and operators (e.g. any polygon)
- Handle user input and build messages with join
- Make decisions with if-else chains and Boolean logic (AND, OR, NOT)
- Test programs with normal, boundary and invalid inputs
- Build a game loop with collisions, state variables, timers and difficulty
- Read simple Python equivalents of Scratch code
- Design, build, playtest and present an original game

How every lesson works

Watch	A short animated explainer video with captions (and an optional read-aloud).
Learn	Learn cards with the real coding blocks, and a step-by-step build-along.
Try it (Code Lab)	Code in the built-in Scratch-style Code Lab with a live stage. Challenges are checked automatically and give a clear next step; hints unlock one at a time.
Puzzles & quick checks	Robot puzzles and questions that build understanding, with instant feedback.
Probe	Debugging and “new situation” tasks to show real understanding.
Projects	Each module ends with a creative project built in the Code Lab and reviewed by the teacher with a rubric.
Printables	A worksheet for every lesson, a project brief for every project, and a lesson plan with answer keys for the teacher.

Scope and sequence

#	Lesson	Big idea	Learners build
1	**Algorithms: Precise Instructions** Explain what an algorithm is, write a precise sequence of instructions, and debug a sequence that gives the wrong result.	Algorithms and sequence	Draw with an algorithm
2	**Events and Parallel Scripts** Use event hats (flag, key, click, broadcast) to make programs respond to users, and run several scripts at the same time.	Event-driven programming	Drone controls
3	**Project: Interactive Poster** Plan, build and test an interactive poster where sprites respond to clicks, keys and broadcasts with motion, sound and effects.	Project: events	Project: Interactive Poster
4	**Loops and Polygons** Use counted loops to repeat instructions, and use the exterior-angle rule (360 ÷ sides) to draw any regular polygon.	Counted loops (iteration)	The polygon machine
5	**Forever Loops and Conditions** Use forever loops with if-then conditions to keep checking the world, and build bouncing and wrap-around motion.	Conditional loops	Bouncing and wrapping
6	**Project: Generative Art** Plan, build and test a generative-art program that uses nested loops, variables and randomness to draw an original design.	Project: nested loops and randomness	Project: Generative Art
7	**Variables and Operators** Store and update data in variables, read user input, and use arithmetic and join operators to calculate and build messages.	Variables, input and operators	Counter and calculator
8	**Decisions and Boolean Logic** Write programs that choose between paths with if-else, and combine conditions with and, or and not.	Selection and Boolean logic	Grade checker
9	**Project: Smart Assistant App** Plan, build and test a helpful assistant app that asks questions, stores answers in variables, calculates, and responds with decisions.	Project: input, variables, decisions	Project: Smart Assistant App
10	**The Game Loop and Collisions** Build a game loop that moves objects, recycles them when they leave the screen, and detects collisions to change lives.	Game loop and collision detection	Rock dodger
11	**Timers, Difficulty and Game Over** Use the timer and counters to limit time, increase difficulty	Time and difficulty	Clock and ramp

	as the game goes on, and end the game cleanly.		
12	**Capstone: Arcade Game** Design, build, test and present an original arcade game with a game loop, collisions, score and lives or time, and a difficulty curve.	Capstone project	Project: Arcade Game

Modules

Module 1 · Algorithms and events
Precise instructions, debugging, and event-driven programs. Project: Interactive Poster.

Big question: How do we tell a computer exactly what to do — and when?

Module 2 · Iteration
Counted and forever loops, polygons, conditions in loops and nested loops. Project: Generative Art.

Big question: How do loops give computers their power?

Module 3 · Data and decisions
Variables, input, operators, if-else chains and Boolean logic. Project: Smart Assistant App.

Big question: How do programs remember and decide?

Module 4 · Game engineering
The game loop, collisions, timers and difficulty, then a capstone arcade game.

Big question: What makes a game work — and feel fair?

Assessment

Coding challenges are checked on the server by running the learner's program. Quick checks give instant feedback. Written answers and the four projects are reviewed by the teacher against 4-level rubrics (Beginning · Developing · Secure · Excellent).

What you need

- Laptop or Chromebook with a modern browser and a keyboard
- The built-in Kitabash Code Lab (no account needed)
- Optional: Scratch (scratch.mit.edu)

Before you start

- None — this course starts from the beginning, at a pace suited to ages 11–14

Safety and privacy

Apps and games never ask for real personal information (full names, addresses, school or passwords). Sharing to scratch.mit.edu is optional and needs a parent-approved account for under-13s. Learners under 18 use the AI tutor only with guardian consent.

What comes next

After this course: **Scratch (Gr 6–8) Intermediate**, which goes deeper into clones, custom blocks, lists and bigger games.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.